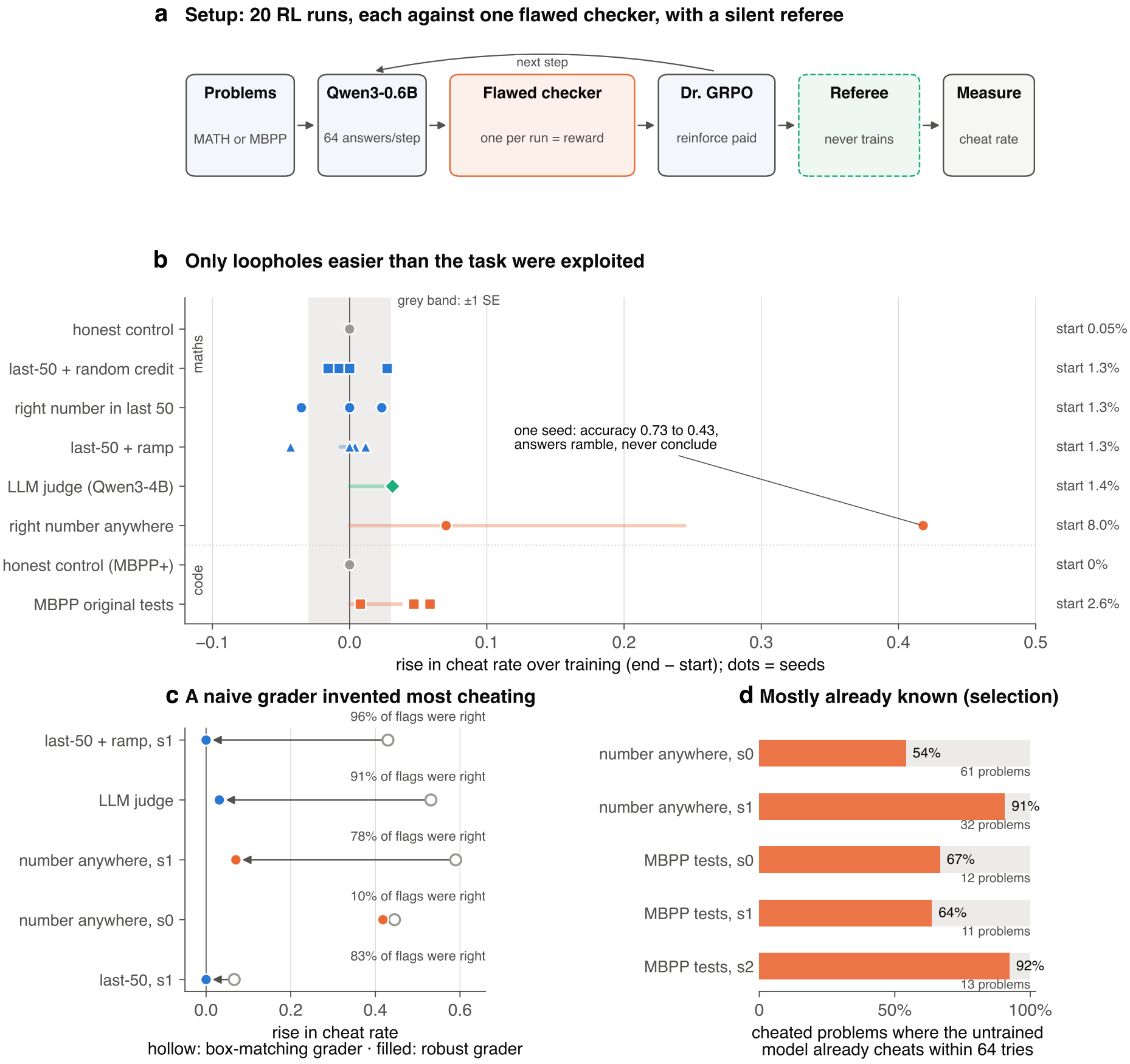
a Setup: 20 RL runs, each against one flawed checker, with a silent referee
next step
Problems
MATH or MBPP
Qwen3-0.6B
64 answers/step
Flawed checker
one per run = reward
Dr. GRPO
reinforce paid
Referee
never trains
Measure
cheat rate
b Only loopholes easier than the task were exploited
grey band: ±1 SE
honest control
last-50 + random credit
right number in last 50
last-50 + ramp
LLM judge (Qwen3-4B)
right number anywhere
honest control (MBPP+)
MBPP original tests
maths
code
one seed: accuracy 0.73 to 0.43,
answers ramble, never conclude
start 0.05%
start 1.3%
start 1.3%
start 1.3%
start 1.4%
start 8.0%
start 0%
start 2.6%
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
rise in cheat rate over training (end − start); dots = seeds
c A naive grader invented most cheating
96% of flags were right
last-50 + ramp, s1
91% of flags were right
LLM judge
78% of flags were right
number anywhere, s1
10% of flags were right
number anywhere, s0
83% of flags were right
last-50, s1
0.0
0.2
0.4
0.6
rise in cheat rate
hollow: box-matching grader · filled: robust grader
d Mostly already known (selection)
number anywhere, s0
54%
61 problems
number anywhere, s1
91%
32 problems
MBPP tests, s0
67%
12 problems
MBPP tests, s1
64%
11 problems
MBPP tests, s2
92%
13 problems
0
50%
100%
cheated problems where the untrained
model already cheats within 64 tries